

Article

Forecasting Trends in Electrical Energy Efficiency in the Food Industry

Saksirin Chinnaket , Pasapitch Chujai Michel  and Pakpoom Chansri 

Division of Electrical Technology Education, King Mongkut's University of Technology Thonburi, Bangkok 10140, Thailand; saksirin.chin@kmutt.ac.th (S.C.); pasapitch.chu@kmutt.ac.th (P.C.M.)

* Correspondence: pakpoom.cha@kmutt.ac.th

Abstract

Trends in electrical energy efficiency are key factors influencing production costs in food industry plants, as all production equipment relies on electricity. Accurate forecasting is essential for predicting future consumption and enabling effective energy management. This study aims to analyze and forecast trends in electrical energy efficiency in the food industry. Production and electricity consumption data from January 2022 to December 2023 were used to calculate the difference in electrical energy (DIFF) and the cumulative sum of electrical energy differences (CUSUM), which served as the basis for forecasting. The Long Short-Term Memory (LSTM) model, based on the deep learning approach, was employed to simulate the algorithmic patterns of electrical energy data in the food industry. Its forecasting performance was then compared with two alternative models, namely decomposition and logistic regression, using evaluation data from January to December 2024. Model accuracy was assessed using the Mean Absolute Percentage Error (MAPE) criterion. The results revealed that the decomposition model achieved lower MAPE values for both DIFF (14.47%) and CUSUM (24.13%), while the logistic regression model yielded higher MAPE values of 73.70% and 66.85%, respectively. Therefore, the decomposition model was identified as the most suitable method for forecasting electrical energy consumption trends in the food industry, providing higher accuracy and reliability than logistic regression. Forecasting energy consumption trends using the decomposition model can support strategic energy planning to enhance efficiency, reduce costs, and promote the sustainable development of the food industry in the future.



Academic Editor: Mihaela Popescu

Received: 21 September 2025

Revised: 17 October 2025

Accepted: 23 October 2025

Published: 29 October 2025

Citation: Chinnaket, S.; Michel, P.C.; Chansri, P. Forecasting Trends in Electrical Energy Efficiency in the Food Industry. *Energies* **2025**, *18*, 5667. <https://doi.org/10.3390/en18215667>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: decomposition; logistic regression; Mean Absolute Percent Error (MAPE); Cumulative Sum Chart (CUSUM); Long Short-Term Memory (LSTM)

1. Introduction

The food industry is among the manufacturing sectors with consistently high electrical energy consumption, particularly in energy-intensive operations such as processing, refrigeration, packaging, and quality control. These processes demand stability and efficiency within the electrical system to ensure optimal energy use in production. Insufficient energy efficiency can result in increased operational costs and energy losses, both in direct production and related auxiliary systems, and presents a significant barrier to long-term sustainable development [1–3]. However, as each stage of the production process relies on machinery [4], which predominantly operates using electrical energy, a stable and continuous power supply is essential. Nevertheless, issues related to low power quality are

commonly encountered in industrial settings. The rising demand for electricity in the industrial sector has therefore emerged as a critical energy concern. Consequently, identifying strategies to reduce electrical energy consumption without compromising production or service quality has become imperative. Systematic monitoring of electricity usage and the analysis of consumption trends play a pivotal role in future energy planning [4], enhancing energy efficiency [5], minimizing operational costs, and mitigating environmental impacts.

Businesses within the industrial sector are placing increasing emphasis on the systematic planning and management of electrical energy, aiming to reduce energy consumption without compromising production or service levels. Monitoring energy usage is essential for analyzing consumption patterns and enhancing energy efficiency. Typically, historical data on production volume and electrical energy consumption are analyzed using statistical methods to gain insights into energy utilization within production processes. This includes exploring correlations between energy consumption and relevant variables, often visualized through scatter plots, and assessing usage trends over time using the Cumulative Sum Chart (CUSUM) method [6–9]. The CUSUM approach is particularly effective in detecting anomalies in electrical energy consumption by visualizing both the differences in energy use and their cumulative effects over time. Such detection facilitates timely intervention and control to optimize energy use in operations. However, despite its strengths in identifying deviations, the CUSUM chart alone does not provide sufficient predictive capability for forecasting future energy efficiency trends. As a result, there is a growing preference within the industrial sector for employing forecasting techniques that can anticipate future electrical energy consumption patterns. Much of the existing literature focuses on retrospective analyses, which serve as foundational data for forecasting models [10]. Effective energy forecasting methods must demonstrate a high degree of accuracy and reliability, as electrical energy consumption is influenced by a wide range of factors, including production volume, seasonal climatic conditions, raw material availability, and broader economic circumstances.

Most accurate electricity consumption forecasting methods employ time series techniques to predict industrial electricity consumption by utilizing historical data to identify patterns for future projections. Various time series forecasting techniques exist, each characterized by distinct analytical properties depending on the nature of the time series data. These techniques often decompose the data into components such as trend, seasonality, cycle, and irregular variations [11–14]. Accurate forecasting requires determining appropriate forecasting horizons—short-term, medium-term, and long-term [15–19]—in order to select the most suitable model [20]. The selection process is typically based on minimizing the mean forecasting error. Furthermore, deep learning models such as Long Short-Term Memory (LSTM) networks, a type of artificial neural network that is capable of learning sequential data and capturing long-term dependencies, have been applied to electricity consumption forecasting and have demonstrated high predictive accuracy [21–23]. Therefore, this research focuses on applying logistic regression and decomposition to compare their accuracy in forecasting trends in electricity efficiency within the food industry [24–26].

This research article aims to study forecasting models of electrical energy consumption efficiency trends in the food industry by employing statistical methods that capture the cumulative difference in electrical energy consumption over 24 months. These historical data are utilized to develop predictive models for future electrical energy efficiency. The analysis compares two forecasting approaches—the logistic regression model and the decomposition model—based on their performance in modeling the DIFF and CUSUM in electrical energy consumption, which serves as an indicator of electrical efficiency trends. Model accuracy is evaluated using the MAPE. The findings from this study may offer

valuable insights for guiding strategic energy planning and improving energy management practices in the industrial sector.

2. Forecasting Methods

2.1. Logistic Regression Method

Logistic regression is a technique used to classify categorical outcomes by estimating the probability that a given observation belongs to a specific group, such as categorizing electricity consumption efficiency as either high or low in the following year. This prediction is based on various explanatory variables, including historical energy consumption, applied technologies, production volume, business size, and other relevant factors. The dependent variable to be predicted must be categorical (e.g., 0 = inefficient; 1 = efficient). Independent variables may include factors such as electricity consumption (kWh), production output (tons), machine age, the presence of automatic control systems, average cold storage temperature, and maintenance frequency, among others. These variables are incorporated into the model to estimate the probability that a given case will fall into one of the predefined categories, as represented in Equation (1).

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}} \quad (1)$$

where $P(Y = 1)$ is the probability that the dependent variable Y equals 1, β_0 is the intercept term, $\beta_{1,2,\dots,n}$ are the coefficients for the independent variables, $X_{1,2,\dots,n}$ are the independent variables, and e is the base of the natural logarithm.

A threshold is established to evaluate electricity usage efficiency, whereby a predicted probability value greater than or equal to 0.5 is classified as “efficient.” The logistic regression model utilizes data from the subsequent year to forecast the category to which a given observation belongs. When applied in this context, logistic regression can support decision-making regarding the prediction of electricity efficiency in the food industry by analyzing the cumulative differences in electricity usage over the preceding two years.

2.2. Decomposition Method

Forecasting the trend in electrical energy efficiency using the decomposition model is a commonly employed technique in time series analysis. This approach involves decomposing time series data into its principal components to facilitate more accurate analysis and forecasting. These components typically include the long-term directional movement in energy efficiency (Trend), recurring patterns observed at regular intervals (Seasonal), cyclical variations that represent irregular yet systematic fluctuations that are often influenced by economic or production cycles (Cyclical), and random variations that cannot be explained by the aforementioned components (Irregular or Residual). In this study, a multiplicative decomposition model is applied, as it is suitable for time series data in which the magnitude of seasonal variations changes proportionally with the level of the trend. The structure of the model is expressed as shown in Equation (2) [11].

$$y_t = S_t T_t \varepsilon_t \quad (2)$$

where y_t is the observed value, S_t is the seasonal component, T_t is the trend effect (sometimes called the trend–cycle effect), and ε_t is the random error component.

In the context of energy efficiency in the food industry, this study utilizes monthly historical data on electricity consumption (kWh) and production volume (tons) over the past five years to compute the cumulative difference in electricity usage. The decomposition method is employed to disaggregate the time series data into its key components, trend,

seasonal, and residual, to assess whether energy efficiency is improving or deteriorating over time. Additionally, the seasonal component is analyzed to identify periods of relatively higher or lower energy consumption. The extracted trend component serves as the basis for forecasting electricity efficiency in the subsequent year.

3. Methodology

The process of forecasting energy consumption trends in the food industry begins with the collection of monthly data on production volume and electrical energy consumption over a 24-month period, from January 2023 to December 2024. The relationship between production volume and electrical energy consumption is then analyzed using linear equations, which serve as reference models for estimating the expected energy consumption based on the actual production levels. The estimated energy consumption values are compared with the actual consumption values to determine the deviations, which are subsequently analyzed using a CUSUM. The cumulative differences obtained from the CUSUM analysis are used to forecast energy efficiency trends for the subsequent 12 months. In addition, as previously reported [27], a Long Short-Term Memory (LSTM) deep learning model is employed in this study. The LSTM network, designed to emulate human memory patterns, processes the CUSUM energy data to identify significant positive and negative deviations within each time period. The algorithm selectively retains crucial information while discarding less relevant CUSUM values, thereby enhancing forecasting accuracy. This time series analysis approach is suitable for sequential data processing and is implemented using Python's Scikit-learn (Python 3.12.0, Scikit-learn 1.7.1) and Time Series Forecasting libraries. This study utilizes the decomposition method, logistic regression, and actual observed data to compare the performance of both models. The forecasting accuracy is evaluated based on the MAPE, as shown in Figure 1.

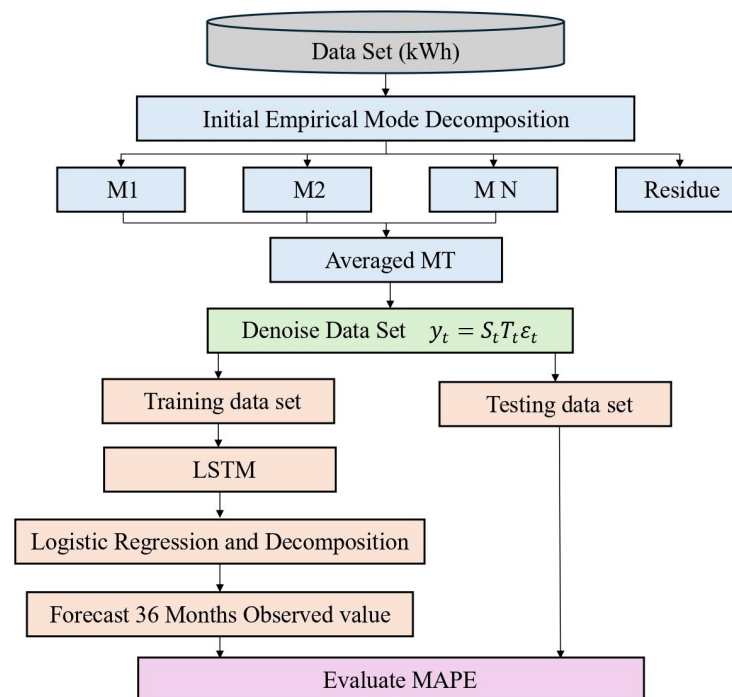


Figure 1. Overview of the data analysis framework applied to forecast electricity consumption in the food industry.

3.1. Cumulative Electricity Consumption Difference Analysis

The analysis of actual electricity usage efficiency can be carried out through an energy monitoring process that utilizes production volume and total electrical energy consumption

data from the food industry over the past 24 months, spanning January 2022 to December 2023. This data is analyzed using statistical tools that are commonly applied in energy management, with the CUSUM being one of the most widely used methods. The CUSUM chart is constructed based on historical data on production volume and electrical energy consumption to monitor and evaluate energy efficiency trends. The process begins with the development of a general linear regression model [28,29], as defined in Equation (3), which is then used to estimate expected electrical energy consumption [30,31]. These expected values serve as a reference for calculating deviations from actual consumption.

$$y = mx + c \quad (3)$$

where y is the quantified electrical energy usage (kWh), x is the monthly production volume (units), m is the gradient (productive dependent energy consumption), and c is the value where the line cuts the y -axis (unproductive energy consumption).

After deriving the estimated amount of electrical energy based on the reference baseline, the next step involves calculating the energy difference and the cumulative energy difference, as shown in Equations (4) and (5).

$$D_i = A_i - y_i \quad (4)$$

$$Q_i = \sum_{i=1}^i D_i \quad (5)$$

where A_i is the actual electrical energy consumed in month i (kWh), y_i is the estimated electrical energy in month i , calculated using the reference linear equation (kWh), D_i is the difference between the actual energy consumed and the estimated energy from the linear equation in month i (kWh), and Q_i is the cumulative sum of energy differences (CUSUM) from month 1 to month i (kWh).

The results of calculating the difference between the expected energy consumption and the actual energy used, based on the cumulative difference (CUSUM) method, are shown in Table 1.

Table 1. Analysis of monthly differences and cumulative differences in electricity consumption over a 24-month period.

YYYY-MM	Production Volume (kg)	Electrical Energy Consumption (kWh)	Baseline-Calculated Electrical Energy Consumption	DIFF	CUSUM
2022-01	100.5	1600.64	1537.17	63.47	63.47
2022-02	101.5	1612.66	1515.6	97.07	160.53
2022-03	91.5	1398.92	1313.82	85.1	182.16
2022-04	101.5	1589.14	1515.6	73.54	158.64
2022-05	105	1602.66	1586.22	16.44	89.98
2022-06	77.5	1197.25	1031.33	165.93	182.37
2022-07	103	1598.69	1545.86	52.83	218.75
2022-08	96	1339.17	1404.62	−65.45	−12.62
2022-09	100.5	1498.69	1495.42	3.27	−62.17
2022-10	101.5	1598.11	1515.6	82.51	85.78
2022-11	101.5	1598.11	1515.6	82.51	85.78
2022-12	101.5	1653.36	1515.6	137.76	122.13
2023-01	105	1621.9	1586.22	35.68	173.44
2023-02	105.5	1605.14	1596.31	8.83	44.51
2023-03	106.5	1598.97	1616.49	−17.52	−8.69
2023-04	98	1499.42	1444.97	54.44	36.92
2023-05	108	1683.67	1646.75	36.91	91.35
2023-06	100.5	1553.36	1495.42	57.94	94.85
2023-07	101.5	1531.11	1515.6	15.51	73.45

Table 1. *Cont.*

YYYY-MM	Production Volume (kg)	Electrical Energy Consumption (kWh)	Baseline-Calculated Electrical Energy Consumption	DIFF	CUSUM
2023-08	101.5	1531.11	1515.6	15.51	73.45
2023-09	101.5	1589.19	1515.6	73.59	−26.01
2023-10	105	1589.36	1586.22	3.14	76.73
2023-11	105.5	1599.11	1596.31	2.8	5.94
2023-12	106.5	1598.69	1616.49	−17.8	−15

3.2. Forecasting and Comparative Analysis of Electricity Usage Efficiency

To forecast the trend of electrical energy efficiency, the DIFF and CUSUM values shown in Table 1 were utilized. Electricity production and consumption data from the canned food industry (Takerng Pineapple Industrial Co., Ltd., Prachuap Khiri Khan, Thailand) were analyzed to assess forecasting performance by comparing two models: the decomposition model and logistic regression. These models employ distinct analytical and forecasting approaches. The decomposition model separates the time series data into trend, seasonal, and residual components, which serve as the foundation for forecasting. Owing to the seasonal characteristics of the data, logistic regression is applied to model the probability of an outcome belonging to a specific group, and the differences in electrical energy are subsequently clustered. The forecasting performance of both models is evaluated using the MAPE, as expressed in Equation (6). The model yielding the lowest MAPE value is considered the most suitable for forecasting electrical energy efficiency [11].

$$MAPE(\%) = \frac{1}{n} \sum_{t=0}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \times 100 \quad (6)$$

where Y_t is the actual value of the data at time t , $\hat{Y}_t$ is the predicted value at time t , and n is the total number of observations.

4. Results and Discussions

This study investigates the forecasting of energy efficiency trends in food industry factories by analysing production volume and electrical energy consumption data. The analysis focuses on calculating the monthly differences in electrical energy consumption and the CUSUM. As shown in Table 1, the results indicate both positive and negative 1energy deviations across individual months, reflecting instances of energy loss or surplus. The dataset used spans from January 2022 to December 2023, with the findings being shown in Figure 2.

Figure 2 shows the CUSUM alongside the DIFF results over a continuous 24-month period. The analysis reveals that the monthly DIFF predominantly indicated positive deviations, with an average energy loss of 83.33%. When examining the CUSUM, the overall trend remained positive, signifying a consistent loss of electrical energy throughout the period. These losses are attributable to both production-related and non-production-related factors. The resulting data from the CUSUM and DIFF analyses were subsequently employed in forecasting models using both the decomposition method and logistic regression to determine the most accurate and suitable approach for predicting future electrical energy consumption efficiency.

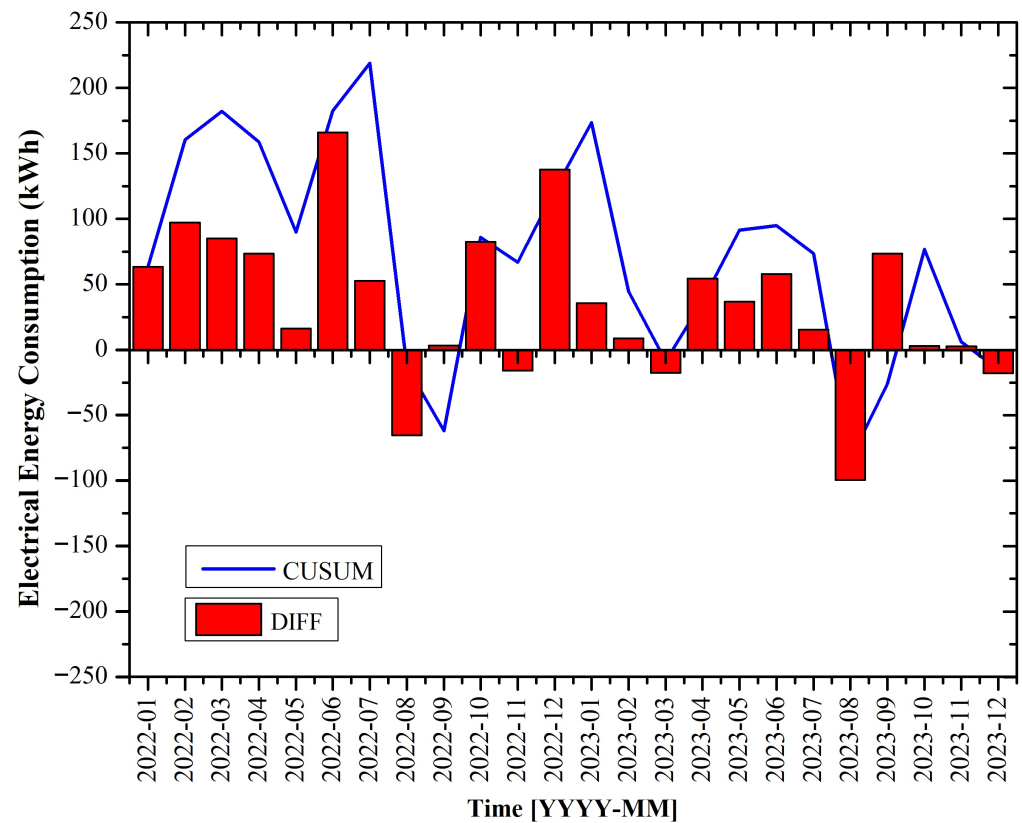


Figure 2. CUSUM and DIFF analysis of electricity consumption over a continuous 24-month period.

In analyzing the forecasting trend of electrical energy consumption using the decomposition and logistic regression models, this study employs a Long Short-Term Memory (LSTM) model based on deep learning techniques to simulate the algorithmic behavior of electrical energy consumption patterns in the food industry. The baseline data on electricity usage shown in Table 1 are utilized to analyze and compare the actual and simulated values generated by the LSTM model, as shown in Figure 3.

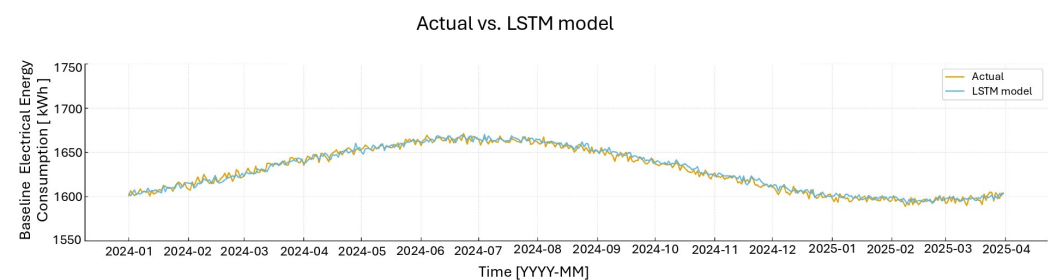


Figure 3. Comparison of the performance of the LSTM model with the actual value.

Figure 3 shows a comparison between the actual daily electrical energy consumption and the forecasted values generated by the LSTM model from January 2024 to March 2025. The forecast line closely overlaps with the actual values for most of the period, demonstrating the model's high accuracy and stability. Electrical energy consumption exhibited an increasing trend from March to August, followed by a decline after September, which aligns with the seasonal production characteristics of the food industry. The LSTM model effectively captured this pattern, achieving a MAPE of 1.84%.

The LSTM model's capability of retaining long-term dependencies through its memory cell architecture enables it to learn temporal variations in energy consumption more effectively than conventional linear models. Consequently, it produces highly accurate

baseline forecasts of electrical energy consumption that reflect trends toward improved energy efficiency. Moreover, the results indicate a gradual downward trend from late 2024 to early 2025, suggesting enhanced energy performance within the manufacturing process. This improvement may be attributed to advancements in energy control systems or increased operational efficiency of factory machinery. Overall, this declining trend represents a positive indicator for future energy demand forecasting.

Figure 4 shows (a) the comparison between the distribution of actual electrical energy consumption and the predicted values obtained from the LSTM model and (b) the distribution of absolute error values. As shown in Figure 4a, the medians of both the actual and predicted data are closely aligned at approximately 1050 kWh, with similar interquartile ranges (IQRs). This indicates that the LSTM model can accurately capture the energy consumption trend, achieving a MAPE of 4.7%, which represents an acceptable level of forecasting accuracy for practical applications in the food industry. In Figure 4b, most of the absolute error values are concentrated below 0.19, with no significant outliers, demonstrating the stability and robustness of the model. The LSTM's capability to retain long-term sequential information allows it to effectively learn energy consumption patterns associated with production cycles and seasonal variations. The results of this analysis suggest that the LSTM model has strong potential for application as a decision-support tool in energy management for food industry facilities, contributing to enhanced energy efficiency and reduced production losses. Subsequently, the LSTM model was applied to forecast the DIFF between the decomposition and logistic regression forecasting models, as illustrated in Figure 5.

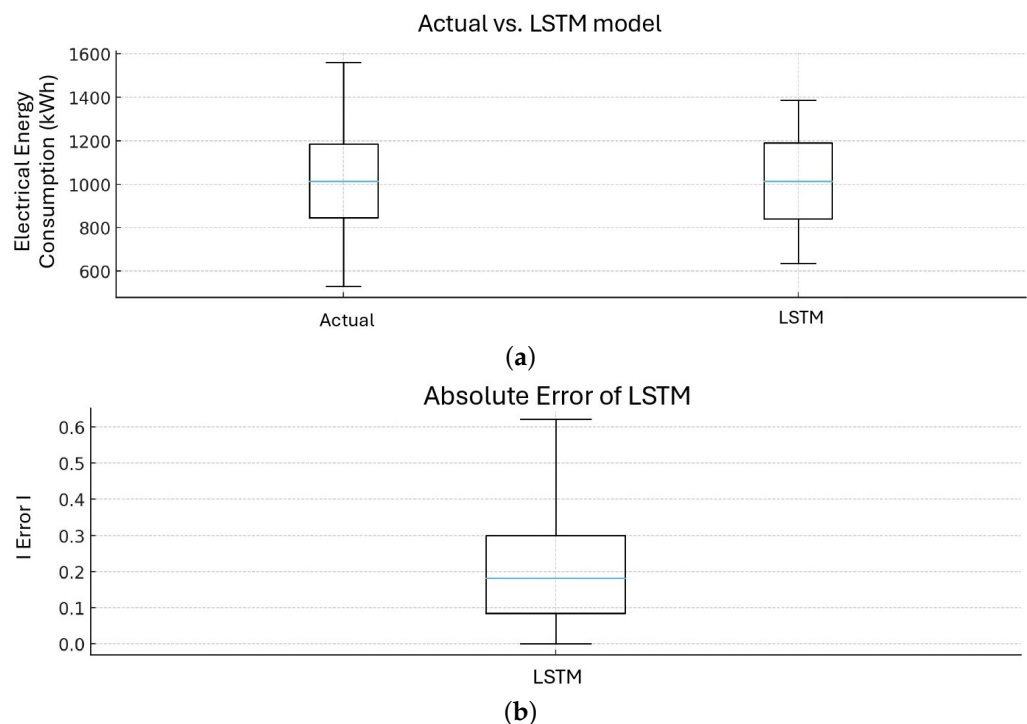


Figure 4. Boxplot comparison and error analysis: (a) actual value vs. predicted LSTM model distributions; (b) absolute error distribution of the LSTM model.

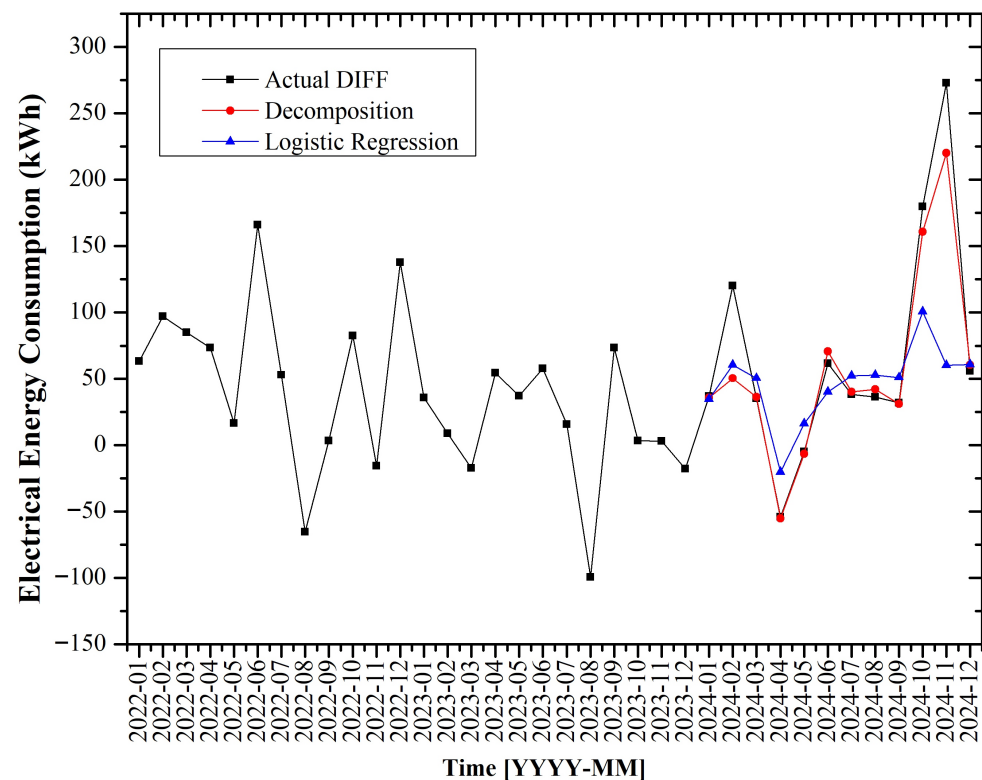


Figure 5. Comparison of forecasted DIFF trends using the decomposition model and logistic regression model.

From Figure 5, which shows the forecasted DIFF values from January to December 2024, it is evident that the decomposition model consistently produces forecasts that are closer to the actual values than those of the logistic regression model. The divergence between the two models becomes particularly pronounced during the final quarter (October to December), where the decomposition model maintains higher accuracy, whereas the logistic regression model consistently underestimates the actual values. These results suggest that the decomposition model provides a more reliable forecast of electrical energy consumption efficiency trends. The discrepancies observed in both models during the early months of 2024 (January to March) may be attributed to overfitting, potentially caused by excessive sensitivity to minor fluctuations in the training data or increased model complexity [32].

From Figure 6, which shows the forecast of the CUSUM trend from January to December 2024, it was observed that the decomposition model yielded predictions that were most closely aligned with the actual trend. This can be attributed to the model's ability to separate trend and seasonal components from residuals, thereby minimizing the influence of short-term anomalies and allowing for a more accurate understanding of temporal patterns. In contrast, the logistic regression model lacks the capability to account for time-dependent structures, which limits its forecasting performance in this context.

To evaluate and compare the accuracy and precision of the forecasting models for electrical energy consumption efficiency trends, the study employed data on monthly differences in electricity usage and their cumulative sums in the food industry sector. The MAPE was used as the evaluation criterion. The model with the lowest MAPE value was deemed the most appropriate and accurate for forecasting electrical energy consumption efficiency, as summarized in Table 2.

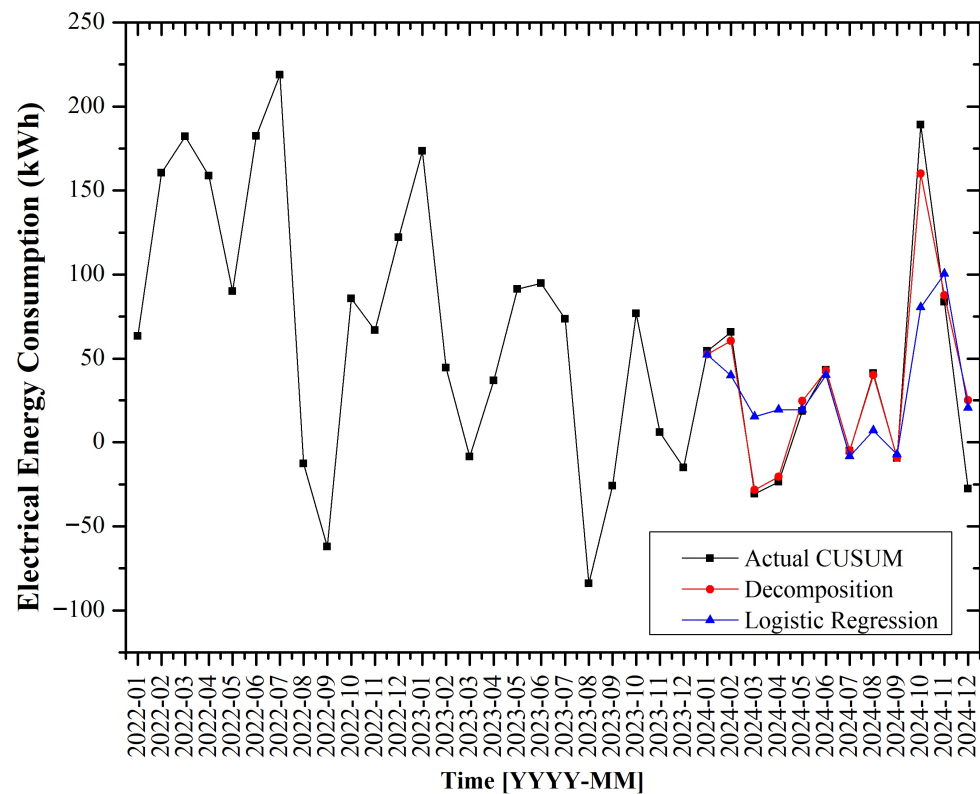


Figure 6. Comparison of forecasted CUSUM trends using the decomposition model and logistic regression model.

Table 2. Comparison of forecasting error (MAPE) between the decomposition model and the logistic regression model.

YYYY-MM	%MAPE (DIFF)		%MAPE (CUSUM)	
	Decomposition	Logistic Regression	Decomposition	Logistic Regression
2024-01	3.00	5.67	3.63	3.82
2024-02	58.12	49.64	7.86	39.08
2024-03	3.49	44.37	7.99	149.87
2024-04	2.08	62.23	13.26	181.83
2024-05	29.43	411.89	33.46	5.04
2024-06	14.57	35.19	1.09	7.03
2024-07	5.31	37.86	7.07	58.70
2024-08	16.56	47.06	2.40	83.04
2024-09	3.11	59.86	2.44	22.59
2024-10	10.55	44.01	15.39	57.43
2024-11	19.26	77.89	4.72	19.96
2024-12	8.19	8.77	190.31	173.87
Average	14.47	73.70	24.13	66.85

Table 2 shows a comparison of the MAPEs of the decomposition and logistic regression models for forecasting DIFF and CUSUM from January to December 2024. The results indicate that the decomposition model consistently yields lower MAPE values for both DIFF and CUSUM, demonstrating its superior ability to capture the underlying trends in electrical energy consumption data more accurately than logistic regression. Notably, the MAPE values for logistic regression show abnormal spikes in certain months, such as May (DIFF = 411.89%) and March (CUSUM = 149.87%), suggesting the model's sensitivity to data fluctuations and potential overfitting during specific periods. In contrast, the decomposition model, which accounts for trends and seasonality, provides more stable and reliable forecasts. These findings underscore the higher forecasting accuracy of the decom-

position model in both DIFF and CUSUM analyses. Therefore, it can be concluded that the decomposition model is the more appropriate and effective approach for forecasting electrical energy consumption efficiency trends in the food industry within the scope of this study.

5. Conclusions

The analysis of electrical energy efficiency trends in food industry plants, based on production volume and electrical energy consumption data from January 2022 to December 2023, revealed an average positive electrical energy loss of 71.63% per month. The CUSUM exhibited a consistently positive trend throughout the period, indicating energy losses originating from both production and non-production activities. The DIFF and CUSUM values were subsequently used to evaluate the forecasting performance of two models: decomposition and logistic regression. The experimental results demonstrated that the decomposition model achieved significantly higher forecasting accuracy than the logistic regression model, yielding a MAPE of 14.47% for DIFF and 24.13% for CUSUM, whereas the logistic regression model recorded average MAPE values of 73.70% for DIFF and 66.85% for CUSUM. Moreover, the logistic regression model exhibited instability and overfitting during certain periods, particularly in months with error rates exceeding 100%, such as March and May. Although logistic regression models based on deep learning using the LSTM algorithm leverage all available data for learning, they are prone to overfitting during specific intervals, resulting in data loss. In contrast, the decomposition model consistently produced trend forecasts that were closely aligned with the actual values, even under conditions of high data volatility. This separate time series analysis ensures reliable statistical results for each period. Therefore, the decomposition model is considered the most appropriate technique for forecasting electrical energy efficiency trends in the food industry. Such forecasting holds substantial engineering value, supporting energy planning, efficiency improvement in manufacturing processes, optimal design of electrical control systems and infrastructure, cost reduction, and the promotion of sustainable production.

Author Contributions: Conceptualization, S.C. and P.C.; methodology, S.C.; software, S.C. and P.C.M.; validation, S.C. and P.C.; formal analysis, S.C. and P.C.; investigation, S.C.; resources, S.C. and P.C.; data curation, P.C.M.; writing—original draft preparation, S.C.; writing—review and editing, S.C. and P.C.; visualization, P.C.; supervision, P.C.; project administration, P.C.; funding acquisition, S.C., P.C.M. and P.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Petch Pra Jom Klao Master's Degree Research Scholarship from King Mongkut's University of Technology Thonburi (KMUTT), under the OF FUNDER grant number 9/2567.

Data Availability Statement: The datasets discussed in this article are not easily accessible due to confidentiality and security regulations that prevent the public dissemination of internal operational data from the corporation.

Acknowledgments: The authors would like to acknowledge the Department of Electrical Engineering, Faculty of Science in Industrial Education, King Mongkut's University of Technology Thonburi, Thailand, for all support in the research. Finally, the authors would like to express their gratitude for the Petch Pra Jom Klao Master's Degree Research Scholarship from King Mongkut's University of Technology Thonburi, which provided this great opportunity and funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Pradella, A.M.; Loures, E.F.R.; Costa, S.E.G.; Lima, E.P. Energy Efficiency in the Food Industry: A Systematic Literature Review. *Arch. Biol. Technol.* **2019**, *62*, e19190002. [[CrossRef](#)]

2. Clairand, J.-M.; Briceño-León, M.; Escrivá-Escrivá, G.; Pantaleo, A.M. Review of Energy Efficiency Technologies in the Food Industry: Trends, Barriers. *IEEE Access* **2020**, *23*, 48015–48029. [\[CrossRef\]](#)
3. Corigliano, O.; Algieri, A. A comprehensive investigation on energy consumptions, impacts, and challenges of the food industry. *Energy Convers. Manag.* **2024**, *23*, 100661. [\[CrossRef\]](#)
4. Pimenov, D.Y.; Der, O.; Patel, G.C.M.; Giasin, K.; Ercetin, A. State-of-the-art review of energy consumption in machining operations: Challenges and trends. *Renew. Sustain. Energy Rev.* **2025**, *224*, 116073. [\[CrossRef\]](#)
5. Charalambous, C.; Polycarpou, A.; Efthymiou, V.; Georghiou, G.E. Enhancing Energy Efficiency and Sustainability with Hybrid AC/DC Microgrids for Energy Transition. In Proceedings of the 3rd International Conference on Energy Transition in the Mediterranean Area (SyNERGY MED), Limassol, Cyprus, 21–23 October 2024; pp. 1–5. [\[CrossRef\]](#)
6. Zheng, S.; Zhang, Y.; Zhou, S.; Ni, Q.; Zuo, J. Comprehensive Energy Consumption Assessment Based on Industry Energy Consumption Structure Part I: Analysis of Energy Consumption in Key Industries. In Proceedings of the IEEE 5th International Electrical and Energy Conference (CIEEC), Nanjing, China, 27–29 May 2022; pp. 4942–4949.
7. Pasiopoulou, I.D.; Vomva, S.A.; Papagiannis, G.K.; Bouhouras, A.S.; Filippidis, S.P.; Christoforidis, G.C. Monitoring the energy behavior of SMEs before and after Energy Efficiency measures: A case study. In Proceedings of the 57th International Universities Power Engineering Conference (UPEC), Istanbul, Turkey, 30 August–2 September 2022; pp. 1–6.
8. Ratsame, P.; Chansri, P. Analysis of Electrical Energy Consumption Efficiency Trends Using Visualization Tools. In Proceedings of the 2024 8th International Conference on Power Energy Systems and Applications (ICoPESA), Hong Kong, 24–26 June 2024; pp. 333–337. [\[CrossRef\]](#)
9. Kongthon, S.; Rakpiamsuk, S.; Rattanawong, W.; Ratsame, P.; Chansri, P. Analysis of Electrical Energy Consumption Efficiency Trends, Case Study: A Textile Industry. In Proceedings of the International Electrical Engineering Congress (iEECON), Hua-Hin, Thailand, 5–7 March 2025.
10. Liu, M.; Mo, C.; Wang, H. Short-term Power Load Forecasting Model Based on CNN-GRU and Dual Attention Mechanism Hybrid Neural Network Model. In Proceedings of the 8th Asia Conference on Power and Electrical Engineering (ACPEE), Tianjin, China, 14–16 April 2023; pp. 1769–1773. [\[CrossRef\]](#)
11. Montgomery, D.C.; Jennings, C.L.; Kulahci, M. *Introduction to Time Series Analysis and Forecasting*, 3rd ed.; Wiley: Hoboken, NJ, USA, 2024.
12. Joseph, M. *Modern Time Series Forecasting with Python*, 2nd ed.; Packt Publishing: Birmingham, UK, 2024.
13. Liu, G.; Xiao, F.; Lin, C.-T.; Cao, Z. A Fuzzy Interval Time-Series Energy and Financial Forecasting Model Using Network-Based Multiple Time-Frequency Spaces and the Induced-Ordered Weighted Averaging Aggregation Operation. *IEEE Trans. Fuzzy Syst.* **2020**, *28*, 2677–2690. [\[CrossRef\]](#)
14. Sangngamseung, N.; Sanarak, A.; Khunwiset, S.; Ratsame, P.; Chansri, P. Comparison Forecasting Methods for Electrical Energy Demand: In Case Study Luggage Manufacturing Industry. In Proceedings of the 13th International Electrical Engineering Congress (iEECON), Hua-Hin, Thailand, 5–7 March 2025; pp. 1–4.
15. Khaled, E.U.; Bayezid, A.A.; Mamun, A.A.; Islam, M.A.; Ahmed, M.R.; Shahabuddin, A.K.M.; Ali, M.M.N. Long Term Electrical Energy Planning Using LEAP: A Case Study for Bangladesh. In Proceedings of the 10th IEEE International Conference on Power Systems (ICPS), Cox’s Bazar, Bangladesh, 13–15 December 2023; pp. 1–6. [\[CrossRef\]](#)
16. Zahari, N.E.M.; Mokhlis, H.; Mansor, N.N.; Sulaima, M.F. Modelling of Hybrid Energy System for Demand-Side Management: A Case Study of Industrial Customer in Malaysia. In Proceedings of the Innovations in Power and Advanced Computing Technologies (i-PACT), Kuala Lumpur, Malaysia, 8–10 December 2023; pp. 1–6. [\[CrossRef\]](#)
17. Podgorelec, V.; Pečnik, Š.; Lahovnik, Z.; Vrbančič, G. Long-term Electricity Consumption Forecasting: How Different Models Forecast into the Future. In Proceedings of the International Conference on Smart Systems and Technologies (SST), Osijek, Croatia, 16–18 October 2024; pp. 139–144. [\[CrossRef\]](#)
18. Hassan, M.H.; Banmongkol, C. Medium Term Load Forecasting for an Industrial Factory Using Bi-LSTM. In Proceedings of the IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia), Pattaya, Thailand, 9–12 July 2024; pp. 259–264. [\[CrossRef\]](#)
19. Wang, Y.; Qian, Y.; Zheng, J.; Hu, Z.; Chen, Q.; Ding, K.; Tian, S. Research on Short-Term Electricity Load Forecasting Model Based on SWT-ResNet-LSTM. In Proceedings of the 3rd International Conference on Energy, Power and Electrical Engineering (EPEE), Wuhan, China, 15–17 September 2023; pp. 1268–1272. [\[CrossRef\]](#)
20. Gao, S.; Li, Y.; Zhong, J. Generation Method for Medium and Long-Term Photovoltaic Power Time Series Considering Variable Order Time Series Characteristics. In Proceedings of the IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia), Chongqing, China, 7–9 July 2023; pp. 2369–2373. [\[CrossRef\]](#)
21. Maarif, M.R.; Saleh, A.R.; Habibi, M.; Fitriyani, N.L.; Syafrudin, M. Energy Usage Forecasting Model Based on Long Short-Term Memory (LSTM) and eXplainable Artificial Intelligence (XAI). *Information* **2023**, *14*, 265. [\[CrossRef\]](#)
22. Abumohsen, M.; Owda, A.Y.; Owda, M. Electrical Load Forecasting Using LSTM, GRU, and RNN Algorithms. *Energies* **2023**, *16*, 2283. [\[CrossRef\]](#)

23. Amalou, I.; Mouhni, N.; Abdali, A. CNN-LSTM architectures for non-stationary time series: Decomposition approach. In Proceedings of the 2024 International Conference on Global Aeronautical Engineering and Satellite Technology (GAST), Marrakesh, Morocco, 24–26 April 2024; pp. 1–5. [\[CrossRef\]](#)
24. Wu, M.-P.; Wu, F. Predicting Residential Electricity Consumption Using CNN-BiLSTM-SA Neural Networks. *IEEE Access* **2024**, *12*, 71555–71565. [\[CrossRef\]](#)
25. Lijuan, Z. Research on Algorithm and Modification of Elman Network Based on Fuzzy Classification and Short-Term Electricity Consumption Prediction. In Proceedings of the IEEE International Conference on Electrical, Automation and Computer Engineering (ICEACE), Changchun, China, 29–31 December 2023; pp. 523–526. [\[CrossRef\]](#)
26. Setiyorini, T.; Frieyadie; Andrianingsih; Safitri, M.; Mardiana, T.; Rahmawati, M. Implementation of Neural Network and Bagging Technique for Predicting Electricity Consumption. In Proceedings of the International Conference on Information Technology Research and Innovation (ICITRI), Jakarta, Indonesia, 10 November 2022; pp. 107–111. [\[CrossRef\]](#)
27. Yang, F.; Fu, X.; Yang, Q.; Chu, Z. Decomposition strategy and attention-based long short-term memory network for multi-step ultra-short-term agricultural power load forecasting. *Expert Syst. Appl.* **2024**, *238*, 122226. [\[CrossRef\]](#)
28. Matjelo, N.J.; Lara, S.P.; Kao, M.; Lepekola, L.; Matobako, M.T.; Singh, M. Differential And Autoregressive Linear Regression Models For Lorentzian Profile Fitting Problem. In Proceedings of the International Conference on Electrical, Computer and Energy Technologies (ICECET), Cape Town, South Africa, 16–17 November 2023; pp. 1–5. [\[CrossRef\]](#)
29. Mohammadi, A.; Chumachenko, D.; Chumachenko, T. Machine Learning Model of COVID-19 Forecasting in Ukraine Based on the Linear Regression. In Proceedings of the IEEE 12th International Conference on Electronics and Information Technologies (ELIT), Lviv, Ukraine, 19–21 May 2021; pp. 149–153. [\[CrossRef\]](#)
30. Peng, B.; Liu, L.; Wang, Y. Monthly electricity consumption forecast of the park based on hybrid forecasting method. In Proceedings of the China International Conference on Electricity Distribution (CICED), Shanghai, China, 7–9 April 2021; pp. 789–793. [\[CrossRef\]](#)
31. Zhang, Q.; Wang, Z.; Wang, W. A prediction Algorithm of Medium and Long Term Electricity Consumption Trend Considering the Characteristics of Industry Electricity Consumption. In Proceedings of the International Conference on Control Science and Electric Power Systems (CSEPS), Shanghai, China, 28–30 May 2021; pp. 67–70. [\[CrossRef\]](#)
32. Jin, L.; Kuang, X.; Huang, H.; Qin, Z.; Wang, Y. Study on the Overfitting of the Artificial Neural Network Forecasting Model. *J. Meteorol. Res.* **2025**, *19*, 216–225.

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.